Basic Statistics

Statistics is a relatively new science with most of the important developments occurring within the last 100 years. Motivation for statistics as a formal scientific discipline came from a need to summarize and draw conclusions from experimental data. For example, Ronald Fisher, Karl Pearson, and Francis Galton each made significant contributions to early statistics in response to their need to analyze experimental agricultural and biological data. Although mathematics is necessary to understand the details of statistical methods, statistics can be thought of as a philosophy for making decisions and drawing conclusions from experimental and observational data. The goal of this unit is to summarize some of the philosophical ideas behind statistics and to provide a general framework in the form of flowcharts for deciding which statistical test is appropriate for a given dataset and scientific question. Also provided is a list of references and software for commonly used statistical methods.

PHILOSOPHY OF STATISTICS

Statistics is a way of thinking about data, more than it is mathematics. Certainly, mathematics is needed for the development and evaluation of statistical methods and for the calculations that are needed to summarize data. However, successful statistical analysis depends on a deeper understanding of why the mathematics is needed and how mathematical results are interpreted and translated into decisions and answers to scientific questions. Here, we review exploratory data analysis, point estimation, and hypothesis testing as three of the most fundamental concepts in statistics. We begin by reviewing the concepts of random variables, populations, random samples, statistics, and parameters.

Random Variables

Random variables are used to describe the possible outcomes of an experiment or observational study. For example, a random variable could be defined as possible blood pressure values for a particular group of subjects. It is standard to denote a random variable with an upper-case letter such as X, Y, or Z, while an actual observed outcome (e.g., a blood pressure value for a single subject) is denoted by a lower-case letter such as x, y, or z. Defining random variables in this way is useful for describing the information being studied. Random variables can be further described by the type of information they represent. For example, blood pressure is measured on a continuous, or quantitative scale, and can thus be classified as a continuous, or quantitative variable. On the other hand, gender and smoking status have only two or several possible outcomes and are thus referred to as discrete, or categorical variables. Knowing the type and number of random variables being studied is important for selecting the appropriate statistical test to use. One goal of statistics is to summarize and explore data in a way that promotes the discovery of interesting patterns. This area of statistics is referred to as exploratory data analysis and is described in more detail below.

Populations and Samples

A very important distinction is made in statistics between the concept of a population and the concept of a sample. A population represents the entire group of possible subjects that could be studied. For example, if it is of interest to know the average blood pressure of people from North America, then the population comprises every single person from that geographic region. In practice, it is usually not possible to take measurements from every single member of a population because of time constraints, technical difficulties, or financial limitations. Thus, as a compromise, we take a random sample of subjects from the population and hope that they are representative of the entire population. Much of statistics depends critically on the size and quality of random samples taken from populations.

Parameters and Statistics

A parameter is a mathematical function of the outcomes of a random variable from the entire population. For example, the average or mean of the blood pressure values from every person in North America is a parameter. As discussed above, we usually study random samples, since it is typically not possible to study the entire population. A function of the outcomes of a random variable from a random sample is called a statistic. Thus, parameters are derived from populations and statistics from samples. It is standard to use Greek letters to represent parameters. For example, mean is usually represented by mu (μ). In contrast,

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statistics are usually represented by the same Greek letter with a ^ symbol (called a hat) over it to signify a statistic. Roman letters are also used occasionally to signify statistics. For example, the Greek letter sigma (σ) is used to represent the parameter for standard deviation while *s* is used for the corresponding statistic.

A central question in statistics is whether sample statistics are accurate estimates of population parameters. This is an area of statistics called point estimation. A second important question in statistics is whether a parameter value is equal to some hypothesized value. This is an area of statistics called hypothesis testing. We discuss both of these areas in more detail below.

Exploratory Data Analysis

The goal of exploratory data analysis is to summarize and visualize data and information in a way that facilitates the identification of trends or interesting patterns that are relevant to the question at hand. As described above, an important concept in statistics is that of random variables. Random variables describe the possible outcomes from an experiment. Much of exploratory data analysis deals with the summary and visualization of random variables. A fundamental exploratory data analysis tool is the histogram, which describes the frequency with which each outcome of the random variable occurs. Histograms are usually plotted as bar plots with the scale of the random variable outcome plotted on the *x* axis and the frequency of the outcome represented by a bar on the *y* axis. Histograms summarize the distribution of the outcomes of the random variable and facilitate the comparison of random variables from different experiments. A classic distribution that is used to describe many biological variables is the normal or Gaussian distribution. Descriptions of histograms and several useful distributions in biology and biomedicine are given by Snedecor and Cochran (1989), Sokal and Rolf (1995), and Rosner (2000). Tufte (1990, 1997, 2001) provides perhaps the best tutorial on visualizing scientific data.

Random variables are also described mathematically by their central tendency and their dispersion. Central tendency is a measure of the center of the distribution and is characterized by the mean or average of the outcomes. When outliers or extreme values are present, the median or midpoint of the values may be a better measure of central tendency. The mode of the data is the most commonly occurring value and may also be useful. Dispersion is a measure of the degree to which outcomes of the random variable differ from their mean and is characterized by the variance or standard deviation. The population variance is the sum of the squared differences between the observed values and their mean divided by the population size. The standard deviation is the square root of the variance and is typically more descriptive and easier to interpret.

In addition to identifying interesting patterns in data, exploratory data analysis is also an important part of testing assumptions of various statistical tests. For example, the identification of extreme observations that are not biologically plausible is important because the inclusion of these "outliers" could dramatically alter the results of a statistical test. We discuss the importance of testing assumptions of statistical tests below.

Point Estimation

When we select a random sample from a population and calculate a statistic from the data, we would like to know that the value of the statistic is close to the value of the true population parameter. For example, we might select a random sample of 1000 subjects from North America and calculate the average blood pressure of subjects in the sample as an estimate of the average blood pressure of all people in the region. How do we know if this is an accurate or precise estimate? Conceptually, we could imagine taking many random samples of 1000 subjects, each time calculating the average. The distribution of these averages would tell us something about the quality of the estimates. For example, do the averages tend to all be very close to one another or are they very spread out? In other words, what is the standard deviation of the averages?

The distribution of statistics from many random samples is called the sampling distribution of the statistic and the standard deviation of the statistics is called the standard error (SE). An important theoretical idea in statistics is that the sampling distribution for the mean will follow a normal or Gaussian distribution regardless of the underlying distribution of the actual data. Furthermore, the center of the sampling distribution will be equal to the true population parameter value. This is referred to as the Central Limit Theorem and forms an important foundation for many statistical tests and ideas. This is described in more detail by Snedecor and Cochran (1989), Sokal and Rolf (1995), and Rosner (2000). The standard error of the mean can be estimated by dividing the standard deviation of the data by the square root of the sample size. The relationship between sample size and standard error is an important one because it says that larger sample sizes will have smaller standard errors. In fact, as the sample size approaches the population size, the values of the sample statistics should become closer and closer to the true population value. When the sample is the population, the statistic and the parameter are the same.

Standard error is a very important measure of the quality of a statistical estimate and can be used to construct confidence intervals that are useful for making precise statements about how far parameter estimates are from the true parameter values that we do not know. A rough approximation of a 95% confidence interval of the mean is an interval constructed from the mean plus or minus twice the standard error. The basic idea of a confidence interval can be best understood in the context of taking multiple random samples from the population. Let us assume that we take many random samples from the population and each time calculate a 95% confidence interval around the estimate of the mean. A 95% confidence interval says that, 95% of the time, the interval constructed will include the true parameter value. Thus, confidence intervals give a statement about the precision of our parameter estimate. More details about the construction of confidence intervals is given by Snedecor and Cochran (1989), Sokal and Rolf (1995), and Rosner (2000). An introduction to the theory of point estimation is given by Rice (1995).

Hypothesis Testing

We have now discussed both exploratory data analysis and point estimation. The final major area of statistics is hypothesis testing. All scientific investigations begin with a motivating question. For example, is the average blood pressure of people from North America different than that of people from South America? From the question, two types of hypotheses are derived. The first is called the null hypothesis. This is generally a theory about the value of one or more population parameters and is the status quo or what is commonly believed or accepted. The alternate hypothesis is generally what you are trying to show. For example, a null hypothesis might be that there is no difference in the mean blood pressure of North Americans and South Americans. The alternate hypothesis might be that North Americans have a higher mean blood pressure than South Americans. It is important to note that statistics cannot prove one or the other hypothesis. Rather, statistics provide evidence from the data that supports one hypothesis or the other.

Much of hypothesis testing is concerned with making decisions about the null and alternate hypotheses. You collect the data, estimate the parameters to be tested, calculate a test statistic that is based on those parameter estimates, and then decide whether the value of the test statistic would be expected if the null hypothesis were true or if the alternate hypothesis were true. Of course, once a decision is made, there is always a chance that you have made the wrong decision. In hypothesis testing, there are two types of errors that can be made. A type I error refers to the situation where the conclusion was made in favor of the alternate hypothesis when the null hypothesis was really true (i.e., a false positive). A type II error refers to the converse situation where the conclusion was made in favor of the null hypothesis when the alternate hypothesis was really true (i.e., a false negative). Thus, a type I error is when you see something that is not there, and a type II error is when you do not see something that is really there. In general, type I errors are thought to be worse than type II errors since you do not want to spend time and resources following up on a finding that is not true.

These statistical decisions are often made by calculating a *p*-value. A *p*-value is simply the probability of observing a test statistic as large, or larger, than the one observed from your data if the null hypothesis were really true. The *p*-values for many test statistics are easily calculated using a computer, thanks to the theoretical work of mathematical statisticians such as Jerzy Neyman. It is common in many statistical analyses to accept a type I error rate of 1 in 20 or 0.05. Thus, the *p*-value calculated from the test statistic must be less than or equal to 0.05 to conclude in favor of the alternate hypothesis. This means there is less than a 1 in 20 chance of making a type I error. However, there is nothing magical about this number. If you are the first to investigate a particular question, you might want to set your significance level higher than 0.05 so that you do not make a type II error. This allows an investigator to generate a working hypothesis that can then be tested in future studies using more stringent criteria. Thus, the significance level used depends on the question context and the investigator's willingness to make type I and/or type II errors.

Prior to carrying out a scientific investigation and a statistical analysis of the resulting data, it is possible to get a feel for your chances of seeing something if it is really there to see. This is referred to as the power of a study and is simply 1 minus the probability of making a type II error. A commonly accepted power for a study is 80% or greater. That is, you would like to know that you have at least an 80% chance of seeing something if it is really there. Increasing the size of the random sample from the population is perhaps the best way to improve the power of a study. The closer your sample is to the true population size, the more likely you are to see something if it is really there. You can also raise your significance level to make it easier to see something. However, this has the potential to increase the type I error rate. More details on hypothesis testing are given by Snedecor and Cochran (1989), Sokal and Rolf (1995), and Rosner (2000). Theoretical ideas are introduced by Rice (1995). Specific details on calculating power for a variety of statistical methods are given by Cohen (1988).

Thus, statistics is a relatively new scientific discipline that uses both mathematics and philosophy for exploratory data analysis, point estimation, and hypothesis testing. The ultimate utility of statistics is for making decisions about hypotheses in order to make inferences (i.e., conclusions) about the answers to scientific questions. In the next section, we provide a guide for selecting the appropriate statistical analysis method.

STATISTICAL METHODS

There are many different statistical methods. Deciding which statistical approach is most appropriate for a particular dataset and scientific question can be very challenging. The purpose of the section is to provide a general guide to selecting some of the most common statistical methods. We have summarized some of the more common statistical approaches in Table A.3M.1. Figures A.3M.1, A.3M.2, and A.3M.3 provide flowcharts for facilitating the decision about which of these tests should be used. Here, we introduce some of the concepts and terminology that are needed to make efficient use of the flowcharts. Finally, we provide a list of commonly used software for each statistical method along with popular references where the details of the approach can be found (Table A.3M.2). It is not the goal of this unit to teach investigators how to do a *t* test, for example. Rather, it is our goal to help each investigator decide which statistic is most appropriate and where to find software and references for facilitating a statistical analysis.

Dependent and Independent Variables

The very first question in each flowchart addresses the number of dependent variables being studied. In statistics, dependent variables are the response variables or the primary endpoint of interest. For example, blood pressure is a dependent variable for investigations that focus on factors that explain interindividual variability in blood pressure. In contrast, independent variables are the explanatory or predictor variables. In the blood pressure example, age could be a continuous independent variable while gender could be a discrete independent variable. Other questions in the flowcharts address either the number or type of independent or dependent variables.

What are you testing?

An important question is whether you are testing the *form* of the relationship between your dependent variable(s) and your independent variable(s) or whether you testing the *degree* of the relationship. When studying the form of the relationship, the specific mathematical relationship between the variables is of particular interest. For example, you may want to know the specific mathematical relationship between dosage of an antihypertensive medication and blood pressure levels in hypertensive subjects. Methods such as linear or logistic regression are useful for this endeavor. However, it is possible to just measure the degree to which the variables are related. For example, it might be of interest to estimate the strength of the relationship between diastolic and systolic blood pressure. Methods such as Pearson's correlation can be used to estimate the degree to which two variables are related.

Assumptions

All statistical tests make certain assumptions about the data or the nature of the question being asked in order for the results to be valid. Most of these assumptions arise directly from the theoretical and mathematical details of the statistic itself. Several of the questions in each flowchart refer to these assumptions. It should be noted that we have tried to list some of the major assumptions for each statistical test. However, a thorough treatment of the assumptions of any specific statistical test should be identified using the references in Table A.3M.2.

An important assumption of many statistical tests is independence. This means it is important that the observations not be related in any way. For example, in genetic studies, siblings or other biologically related relatives are not

Table A.3M.1 Short Descriptions of Each Statistical Method Presented in Figures A.3M.1-A.3M.3

Statistic[a]	Descriptions
ANCOVA	Used to test the form of relationship by comparing the means of a normally distributed, homoscedastic, continuous, dependent variable between groups within the discrete, independent variable, while controlling for one or more continuous confounding variables, or covariates.
ANOVA	Used to test the form of relationship by comparing the means of a normally distributed, homoscedastic, continuous, dependent variable across an arbitrary number of discrete groups within the independent variable.
Canonical correlation	Quantifies the degree of relationship between a set of independent variables, which are discrete, continuous or both, and a set of discrete and/or continuous, dependent variables.
Chi-square test of independence	Used to examine the form of relationship between two discrete variables and to determine if the relationship observed is significantly different from what is expected if there were no relationship between the variables.
Fisher's exact test	A nonparametric alternative to the Chi-square test.
Kruskal-Wallis	A nonparametric alternative to ANOVA.
Linear discriminant analysis	Used to predict membership among groups within the discrete, dependent variable across one or more continuous, independent variables.
Linear regression	Determines the form of relationship between a normally distributed, homoscedastic, continuous, dependent variable and one or more discrete or continuous independent variables.
Logistic regression	Determines the form relationship between a discrete dependent variable and one or more discrete and/or continuous, independent variables.
MANCOVA	Used to test the form of relationship by comparing the means of two or more normally distributed, homoscedastic, continuous, dependent variables across groups within the discrete, independent variable, while controlling for one or more continuous, confounding variables or covariates.
Mann-Whitney U test	A nonparametric alternative to the two-sample *t* test.
MANOVA	Used to test the form of relationship by comparing the means of two or more normally distributed, homoscedastic, continuous, dependent variables across an arbitrary number of groups within the discrete, independent variable.
McNemar's test	A nonparametric alternative to the Chi-square test.
One Sample *t* test	Used to test the form of relationship by determining if the mean of a normally distributed, homoscedastic, continuous, dependent variable is significantly different from the hypothesized value.
Paired *t* test	Used to test the form of relationship by determining if the mean of a normally distributed, homoscedastic, continuous, dependent variable is significantly different between two related groups within the discrete, independent variable.
Pearson correlation	Quantifies the degree of relationship between two normally distributed, homoscedastic, continuous variables.
Repeated measures ANOVA	Used to test the form of relationship by comparing means of a normally distributed, homoscedastic, continuous, dependent variable measured across different time points within the discrete, independent variable.
Spearman's rank correlation	A nonparametric alternative to the Pearson correlation.
Two-sample *t* test	Used to test the form of relationship by determining if the mean of a normally distributed, homoscedastic, continuous, dependent variable is significantly different between two independent groups within the discrete, independent variable.
Wilcoxon paired samples test	A nonparametric alternative to the paired *t* test.

[a]ANCOVA, analysis of covariance; ANOVA, analysis of variance; MANCOVA, multivariate analysis of covariance; MANOVA, multivariate analysis of variance.

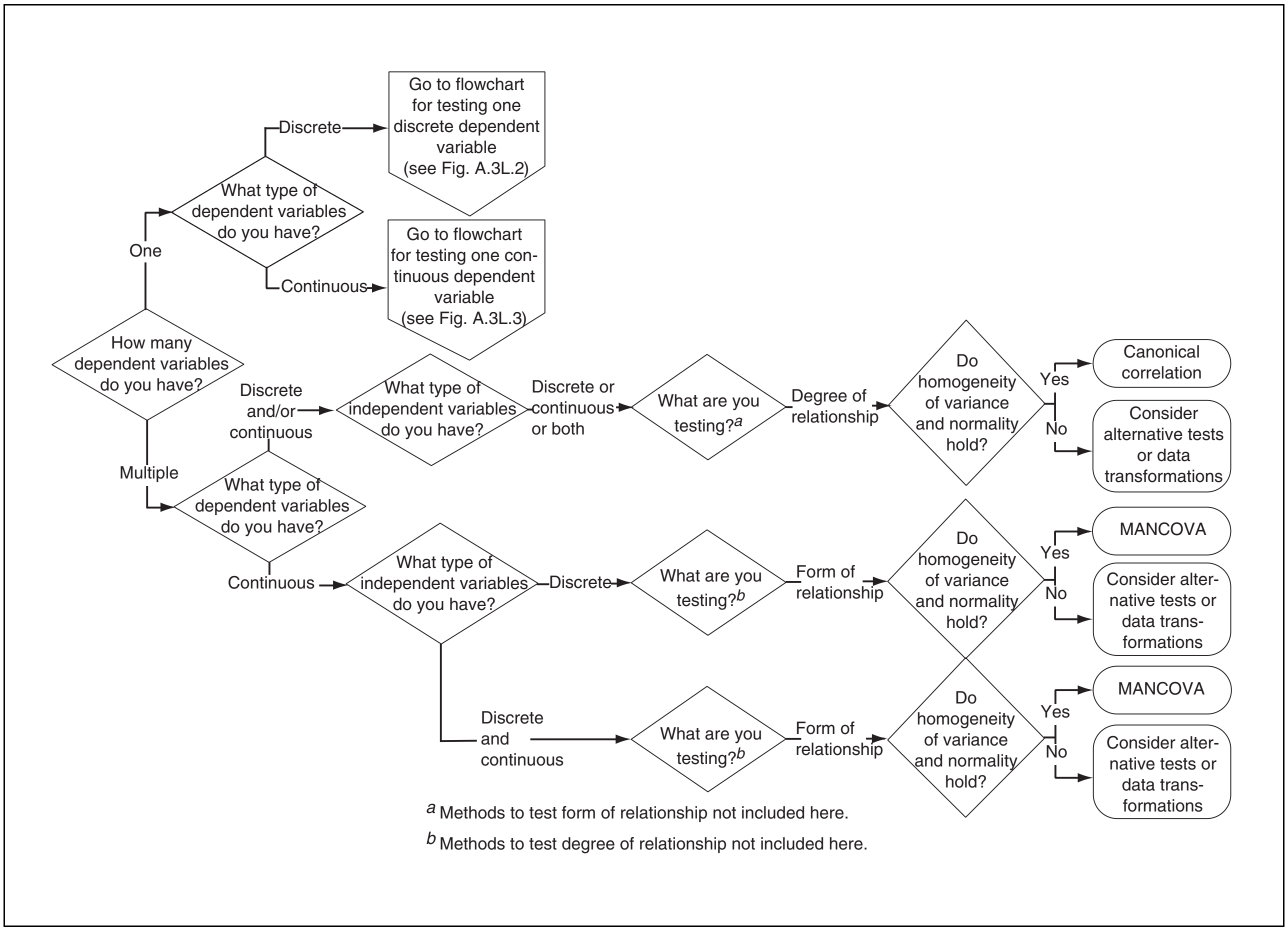

Figure A.3M.1 The first of three flowcharts for deciding which statistical method is most appropriate for a given question and dataset. This flowchart summarizes statistical methods that are useful when multiple discrete and/or continuous dependent variables are studied.

independent since they share common ancestors from whom their DNA was inherited. A statistical test such as McNemar's test that is not sensitive to a lack of independence must be used in these situations.

Another important assumption of many statistical tests is normality of the data and homogeneity of variance. Methods such as linear regression and the *t* test are particularly sensitive to these assumptions. For these assumptions, it is important to formally evaluate whether the dependent variable follows a normal distribution and whether the variance of the dependent variable differs at different levels of the independent variables. Most of the references listed in Table A.3M.2 provide information about testing these assumptions. If the assumptions are violated, there are several options. Sometimes it is possible to apply a mathematical transformation such as the log function to the dependent variable to adjust the data to normality. An alternative is to use nonparametric statistics that are not sensitive to violations of these particular assumptions (Hollander and Wolfe, 1999). Some of these statistics such as Spearman's rank correlation require ranking the observations of the variables and using the ranks in the statistical analysis.

Statistical Software

There are many different software packages that are available for conducting statistical analyses. These all differ according to the particular statistics available, the price, the operating system, and the ease of use. In Table A.3M.2, we have selected several of the more popular statistical software packages and indicated whether their coverage includes each statistical method in Table A.3M.1. The software packages listed include Excel, R, SAS, S-Plus, SPSS, Stata, and Statistica. These are all commercially available except R, which is freely distributed (*http://www.r-project.org/*). Although R is free, it is also the most technically

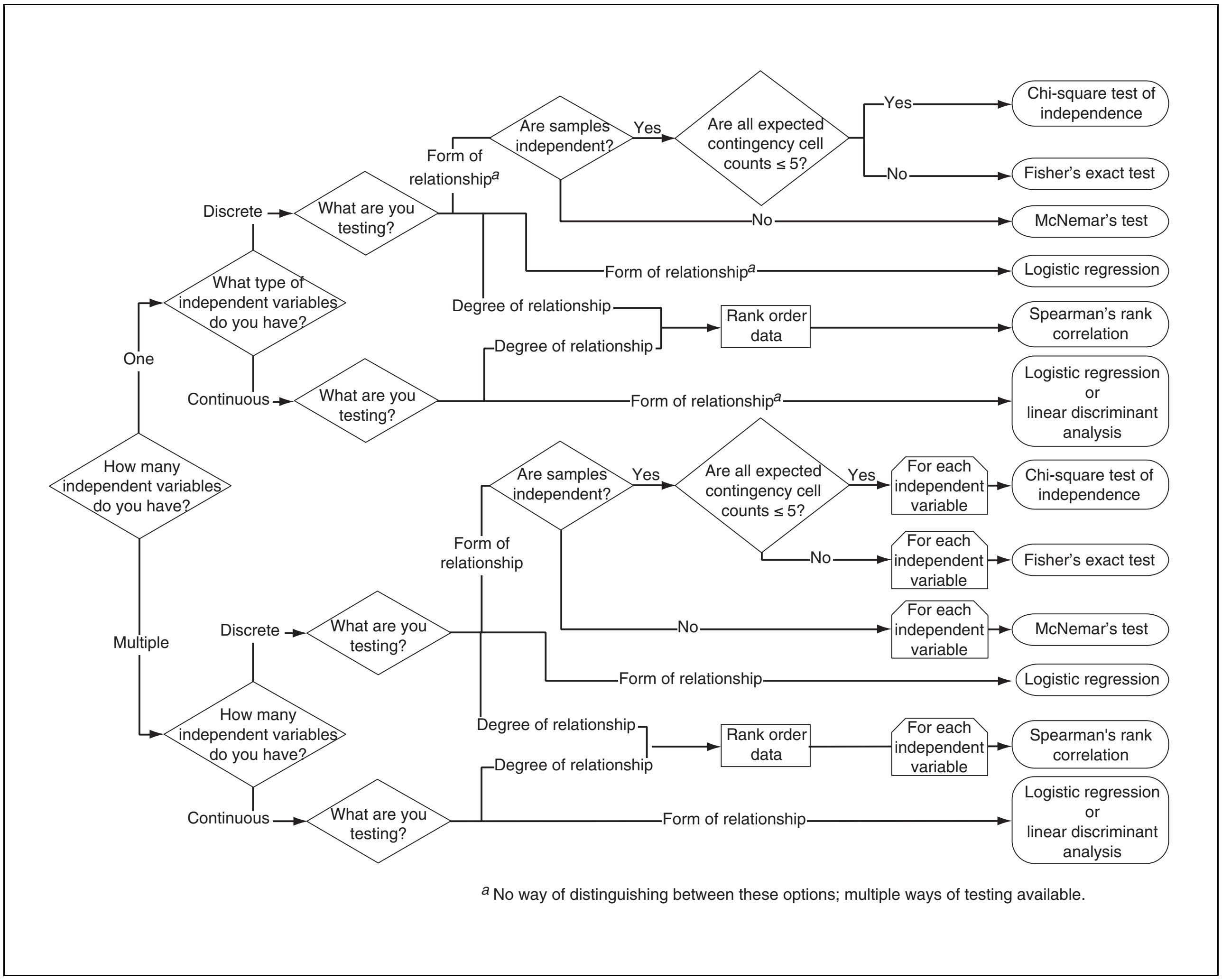

Figure A.3M.2 The second of three flowcharts for deciding which statistical method is most appropriate for a given question and dataset. This flowchart summarizes statistical methods that are useful when one discrete dependent variable is studied.

challenging to use since it does not include a nice graphical user interface (GUI). It should be noted that many of the available software packages include a programming language that makes it possible to develop additional statistical methods.

Statistics References

As with the software, there are many good references for statistical methods. Below, we have selected several general and specific references and have indicated each of the statistical methods covered in those texts. These were selected because they are clearly written and/or are considered to be the authoritative text in the field. The book by Snedecor and Cochran (1989) is a general text that covers many fundamental concepts and methods in statistics. The texts by Rosner (2000) and Sokal and Rohlf (1995) are general texts that cover statistical concepts and methods specifically for the biological and biomedical sciences. The text by Agresti (2002) has a central focus on statistics for categorical or discrete data. The Neter et al. (1990) text covers methods such as ANOVA and linear regression. The text by Hollander and Wolfe (1999) focuses almost exclusively on nonparametric statistics while the Hosmer and Lemeshow (2000) text focuses almost exclusively on logistic regression methods. The Tabachnick and Fidell (1996) text focuses primarily on multivariate statistics that are used when there are multiple dependent and/or independent variables. We anticipate that these texts will be a useful starting point for learning more about the statistical concepts and specific statistical tests mentioned here.

Example

The following example was taken from page 321 of Rosner (2000) to illustrate the use of the flowcharts included with this unit (see Fig

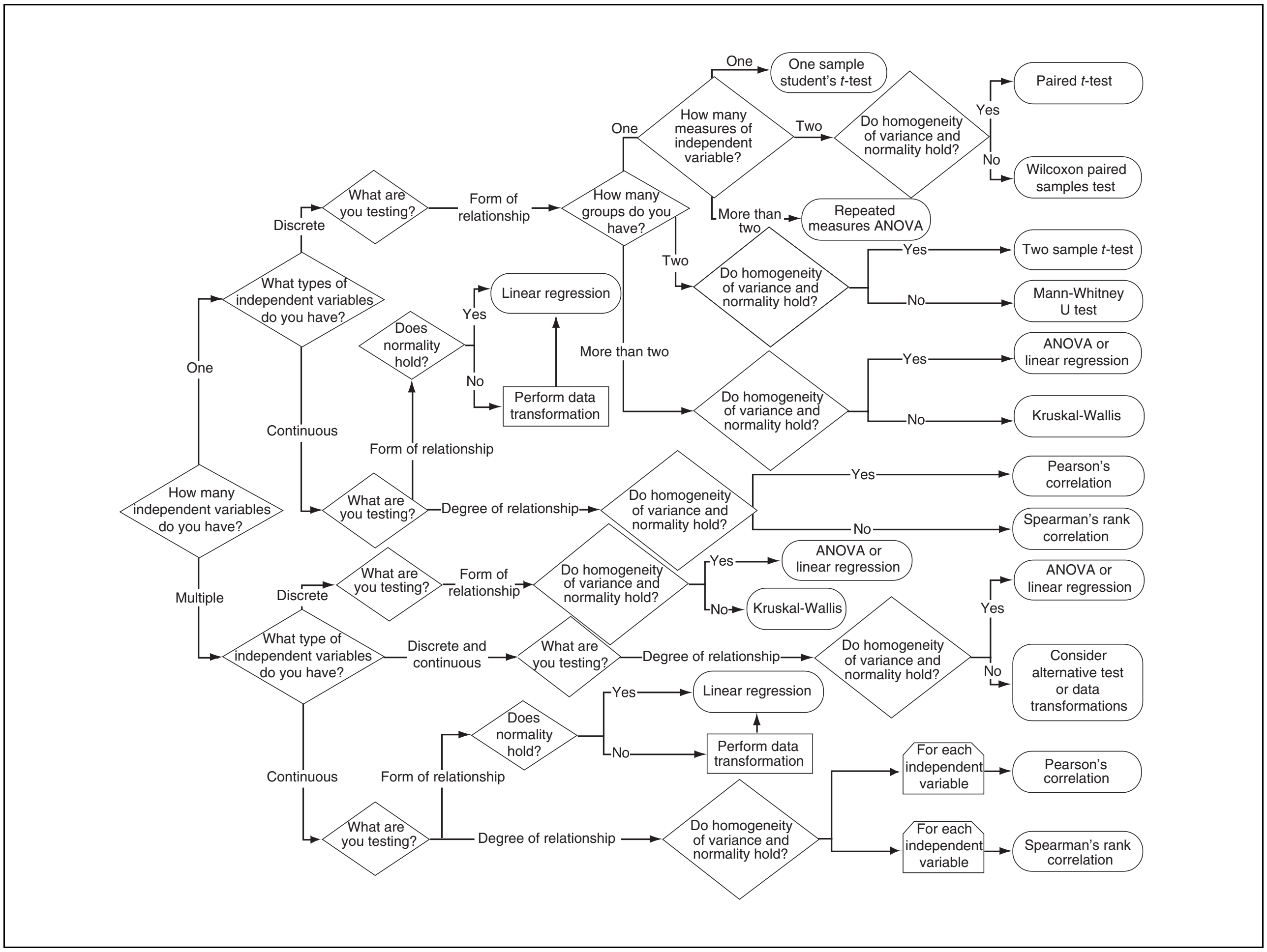

Figure A.3M.3 The third of three flowcharts for deciding which statistical method is most appropriate for a given question and dataset. This flowchart summarizes statistical methods that are useful when one continuous dependent variable is studied.

A.3M.1, Fig. A.3M.2, and Fig. A.3M.3). In this example, visual field area (VFA) was measured in eight unrelated patients that have retinitis pigmentosa (RP) and a point mutation in the rhodopsin gene. In addition, VFA was measured in 140 unrelated RP patients that do not have the rhodopsin mutation. A scientific question might be whether mean VFA is different between RP patients with and without the mutation. The data collected indicate an estimated mean VFA of 7.11 for those with the mutation and 7.99 for those without. The corresponding standard deviations are 1.21 and 1.32, respectively. In Figure A.3M.1, the first question asked is about the number of dependent variables. In this example, VFA is the only dependent variable, since this is the outcome of interest. The next question asked is about type of dependent variable. The VFA variable is a continuous one since it can be measured on finer and finer scales. This leads to the flowchart in Figure A.3M.3. The first question in that flowchart is about the number of independent variables. Here, the only independent variable is rhodopsin gene mutation status. This leads to a question about the type of independent variable. Since there are only two possible outcomes, rhodopsin mutation is a discrete variable. Thus, there are two possible groups or levels for the variable, which answers the next question. The final question concerns the assumptions for the statistical test that this line of questions leads to. The first assumption is homogeneity of variance. That is, is the variance of the dependent variable the same among the two groups? The second assumption is that the dependent variable is normally distributed. Both of these assumptions must first be tested before the appropriate statistical approach is selected. Senedecor and Cochran (1989) and Sokal and Rohlf (1995) have good descriptions about how to carry out these tests. Many of the software packages will perform these tests as well. If the assumptions hold, then the two-sample *t* test should be used. If one or both of the assumptions do not hold, there are two

Table A.3M.2 Software and References for Each Statistical Method Presented in Figures A.3M.1-A.3M.3

Statistic[a]	Software	References
ANCOVA	SAS, SPSS, Stata	Neter et al., 1990; Rosner, 2000; Snedecor and Cochran, 1989; Sokal and Rohlf, 1995; Tabachnick and Fidell, 1996
ANOVA	Excel, R, SAS, S-PLUS, SPSS, Stata, Statistica	Agresti, 2002; Neter et al., 1990; Rosner, 2000; Snedecor and Cochran, 1989; Sokal and Rohlf, 1995; Tabachnick and Fidell, 1996
Canonical correlation	R, SAS, S-PLUS, SPSS, Stata, Statistica	Agresti, 2002; Tabachnick and Fidell, 1996
Chi-square test of independence	Excel, R, SAS, S-PLUS, SPSS, Stata	Agresti, 2002; Hollander and Wolfe, 1999; Rosner, 2000; Snedecor and Cochran, 1989; Sokal and Rohlf, 1995; Tabachnick and Fidell, 1996
Fisher's exact test	R, SAS, S-PLUS, SPSS, Stata, Statistica	Agresti, 2002; Hollander and Wolfe, 1999; Rosner, 2000; Sokal and Rohlf, 1995
Kruskal-Wallis	R, SAS, S-PLUS, SPSS, Stata	Hollander and Wolfe, 1999; Neter et al., 1990; Rosner, 2000; Sokal and Rohlf, 1995
Linear discriminant analysis	SAS, SPSS, Stata	Tabachnick and Fidell, 1996
Linear regression	Excel, R, SAS, S-PLUS, SPSS, Stata, Statistica	Hollander and Wolfe, 1999; Neter et al., 1990; Rosner, 2000; Snedecor and Cochran, 1989; Sokal and Rohlf, 1995; Tabachnick and Fidell, 1996
Logistic regression	R, SAS, S-PLUS, SPSS, Stata	Agresti, 2002; Hosmer and Lemeshow, 2000; Neter et al., 1990; Rosner, 2000; Sokal and Rohlf, 1995; Tabachnick and Fidell, 1996
MANCOVA	SAS, SPSS, Stata	Rosner, 2000; Snedecor and Cochran, 1989; Tabachnick and Fidell, 1996
Mann-Whitney U test	Excel, SAS, SPSS, Stata	Hollander and Wolfe, 1999; Rosner, 2000; Snedecor and Cochran, 1989; Sokal and Rohlf, 1995
MANOVA	SAS, S-PLUS, SPSS, Stata, Statistica	Rosner, 2000; Snedecor and Cochran, 1989; Sokal and Rohlf, 1995; Tabachnick and Fidell, 1996
McNemar's test	R, SAS, S-PLUS, SPSS	Agresti, 2002; Hollander and Wolfe, 1999; Rosner, 2000; Sokal and Rohlf, 1995; Tabachnick and Fidell, 1996
One-Sample *t*test	Excel, R, SAS, S-PLUS, SPSS, Stata	Rosner, 2000; Snedecor and Cochran, 1989; Sokal and Rohlf, 1995
Paired *t* test	Excel, R, SAS, S-PLUS, SPSS, Stata	Rosner, 2000; Snedecor and Cochran, 1989; Sokal and Rohlf, 1995
Pearson correlation	Excel, R, SAS, S-PLUS, SPSS, Stata, Statistica	Hollander and Wolfe, 1999; Rosner, 2000; Snedecor and Cochran, 1989; Sokal and Rohlf, 1995; Tabachnick and Fidell, 1996
Repeated measures ANOVA	Excel, SAS, SPSS, Stata	Neter et al., 1990; Snedecor and Cochran, 1989; Sokal and Rohlf, 1995; Tabachnick and Fidell, 1996
Spearman's rank correlation	SAS, SPSS, Stata	Hollander and Wolfe, 1999; Rosner, 2000; Sokal and Rohlf, 1995
Two-sample *t* test	Excel, R, SAS, S-PLUS, SPSS, Stata	Rosner, 2000; Snedecor and Cochran, 1989; Sokal and Rohlf, 1995

[a]ANCOVA, analysis of covariance; ANOVA, analysis of variance; MANCOVA, multivariate analysis of covariance; MANOVA, multivariate analysis of variance.

choices. First, the data can be transformed and then reevaluated. If the assumptions hold after transformation, then the *t* test can be used. The alternative is to use a nonparametric test such as the Mann-Whitney U test that is not sensitive to these particular assumptions.

SUMMARY

In this unit, we have summarized the philosophical underpinnings of statistics, which should facilitate the data analysis and decision-making process. We have provided explanations of the elemental terms used in statistics and have outlined the key concepts in exploratory data analysis, point estimation, and hypothesis testing. We anticipate that this brief introduction will provide the foundation necessary to better understand the field of statistics and to decide which statistical test is most appropriate for a given question and dataset. It is also anticipated that this unit will provide the necessary starting point for identifying additional references and resources for additional details about specific statistical methods and concepts.

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